Ashwathram Prathap

prathapashwathram@gmail.com | linkedin.com/in/ashwathram-prathap | ashwathram.ca

Education

University of Waterloo — Waterloo, ON
Bachelor of Computer Science, Co-op (Business & Computer Science Double Degree) · Member, WARG

Wilfrid Laurier University — Waterloo, ON
Honours Bachelor of Business Administration (BBA)

Experience

Social Media & Marketing Consultant — 2025 – 2026
Zebra Canada — *Canada*
- Optimized ad targeting and campaign performance by analyzing audience segmentation, A/B test, and ROAS data
- Established a venture delivering Gen Z-focused campaigns to multinational medium-to-large business clients
- Produced branded posts, reels, and short-form video for a corporate marketing team across platforms

Policy & Communications Intern — 2025
Office of the Member of Parliament, Constituency Office — *Milton, ON*
- Drafted a federal budget request for $4M annually over five years for environmental rehabilitation and bylaw enforcement
- Analyzed traffic, emissions, and labour impacts of a contested CN logistics hub for a federal mitigation plan
- Created 30+ social media posts and consulted community stakeholders on youth-employment outreach

Taekwondo Instructor (3rd Dan Black Belt) — 2021 – 2024
Top Level Taekwondo & Martial Arts Academy
- Taught students ages 5 to 50, from one-on-one sessions to 30-person classes, with character education
- Adapted lessons for beginner to advanced students, including students with special needs

Projects

Sidecall: PE Portfolio Breach Response | *Cal Hacks 13.0, Convex, React, TypeScript, Claude* — 2026
- Built with a team of 4; flags portfolio companies sharing a breached vendor login and holds at-risk payments
- Payments stay held until a controller verifies them; holds are rule-based, with Claude only drafting alerts
- 0 misses and 0 false holds in 18 test scenarios; one report holds 304 payments across 303 companies in 1.4 s

LLM Inference Engine from Scratch | *C++20, Apple Metal, Codex* — In Progress
- Building an LLM inference engine from scratch, directing Codex AI coding agents through a four-milestone plan
- Ran GPT-2 (124M) end to end on CPU, matching a Hugging Face reference, at an 11.23 tokens/s naive baseline
- Next milestones: Qwen2.5 chat, a Metal GPU backend, and 4-bit quantization benchmarked against llama.cpp

PID-Tuned Reaction-Wheel Balancing Cube | *Arduino C++, IMU* — In Progress
- Co-building a cube that balances on its corner using reaction wheels, as a step toward a scratch-built UAV
- Wrote the control code (IMU reads, quaternion math, PID loop) while my partner did CAD; learning PCB design

UniMate: AI Planning Assistant | *TypeScript, Node.js, Snowflake Cortex, ElevenLabs* — 2026
- Won the Snowflake Design and Blackboard AI Design awards at Pivot Hacks; built in 12 hours with a team of 4
- Personally built the Chrome extension; the app uses Cortex to turn course pages into prioritized daily actions

Leadership

Founder & President, Student Business & Finance Association — 2024 – 2026
Bishop P. F. Reding CSS — *Oakville & Milton, ON*
- Founded a business association after pitching every business teacher; scaled it to multiple 30+ member chapters
- Led a 3-person exec team running pitching workshops; 5 members went on to major school leadership roles

Team Lead & Business Lead, FIRST Robotics Competition (FRC) — 2024 – 2026
Bishop P. F. Reding CSS — *Milton, ON*
- Raised over $100,000 in funding alongside teammates, sustaining the team's build, travel, and competitions
- Ran a 12-student business team: sponsor outreach, cold-calling training, and in-kind sponsorships
- Led the full team and directed development of the robot's autonomous driving routines

Lead Senator & Human Rights Council Chair — 2024 – 2026
HCDSB Student Senate — *Halton Catholic DSB*
- Chaired a 36,000-student board's Human Rights & Equity Council; worked with 12+ senators to upgrade special ed

Skills & Activities

Programming: Java, TypeScript, Node.js, C++ (Arduino), Python (in progress)
AI Tools: Claude Code, Codex, Snowflake Cortex, ElevenLabs
Business: Excel (Financial Modelling), Power BI, Microsoft Suite, G Suite, Canva, Video Editing
Certifications: Government of Canada Reliability Clearance (2025), Bronze Medallion (2026)
Activities: 2x DECA Regional Champion, Cross Country Regional Champion, triple jump, cricket · English, Tamil